Slope-intercept form

1 b stands for y-intercept and m stands for slope.

2

a Slope $= -\frac{4}{3}$, y-intercept $= -7$

b Slope $= 6$, y-intercept $= 9$

c Slope $= -1$, y-intercept $= 0$

d Slope $= -1$, y-intercept $= -8$

3

a

i $y = 15x - 10$

ii Slope $= 15$, y-intercept $= -10$

b

i $y = 4x + 3$

ii Slope $= 4$, y-intercept $= 3$

4

a $-\frac{1}{9}$

b $-\frac{1}{4}$

5

a $y = -8x - 9$

b $y = -5x + 7$

c $y = \frac{2x}{3} + 2$

d $y = -6x - 7$

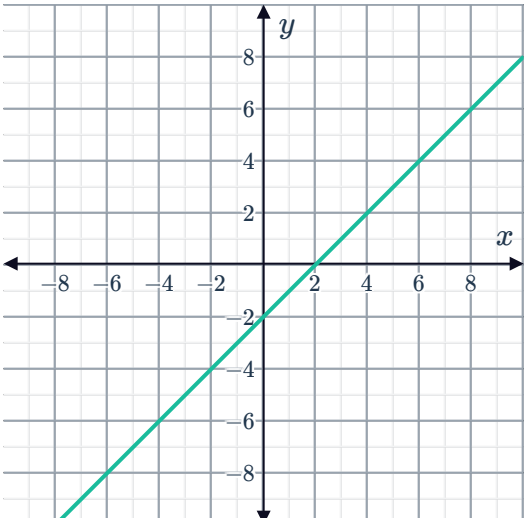
e $y = \frac{2}{13}$

6

a 9

b $y = 9x - 1$

7



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8

a

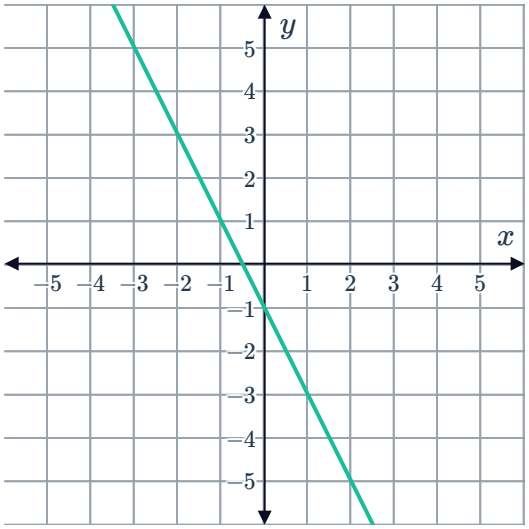
i $\frac{2}{5}$ ii 2 iii $y = \frac{2x}{5} + 2$

b

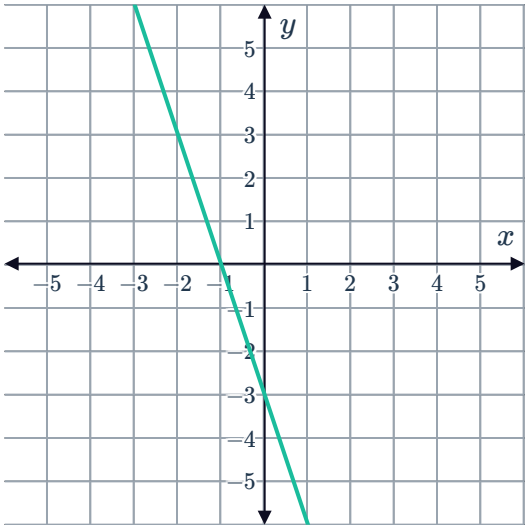
i 3 ii -1 iii $y = 3x - 1$

9

a



b



10

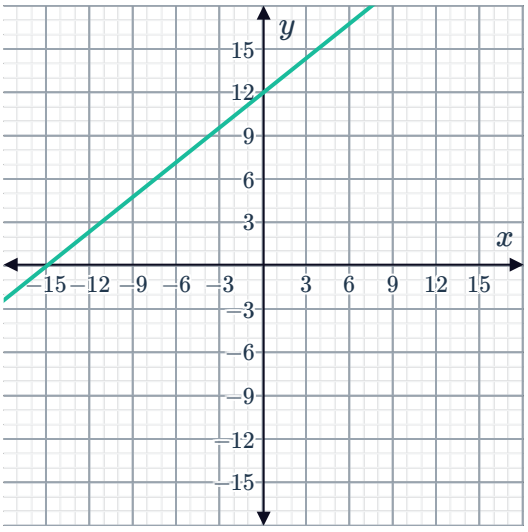
a $b = -23$ **b** $y = -3x - 23$

11

a

i $b = 12$ ii $y = \frac{4x}{5} + 12$

iii $(-15, 0)$ iv



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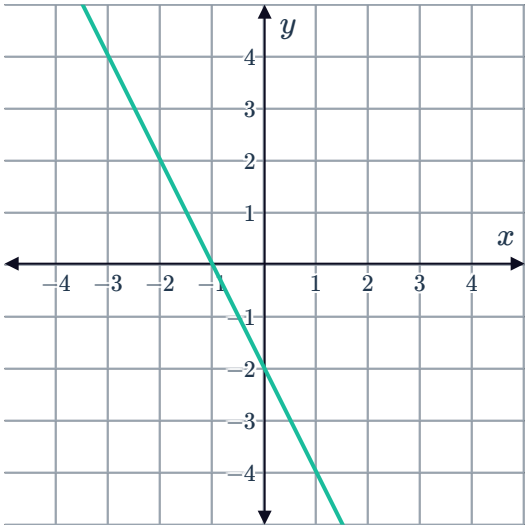
b

i $b = -2$

ii $y = -2x - 2$

iii $(-1, 0)$

iv



12

a $y = -x + 2$

b $a = 8$

13

a $y = 1 + 8x$

b

x	0	1	2
y	-3	2	7

14 $y + 6x = 2$

15

x	0	1	2
y	3	7	11

16 Larger

17 Smaller

18 More rapidly

19 **A**

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20 Smaller

21 Larger

22

a $60\ \text{km/h}$ **b** $71\ \text{km/h}$

c Bus

23

a $u = 20$ **b** Compunet

24

a $4\ \text{mg/min}$ **b** $5.9\ \text{mg/min}$

c In capsule form

Additional questions

25 Line 1: $y = 2 + 6x$
Line 3: $y = 6x$
Line 6: $y = 6x - 2$

26

a Yes, $(0, 2)$ **b** Yes, $(0, 0)$ **c** Yes, $(0, 1)$ **d** No

27

a Yes **b** No **c** Yes **d** No

28

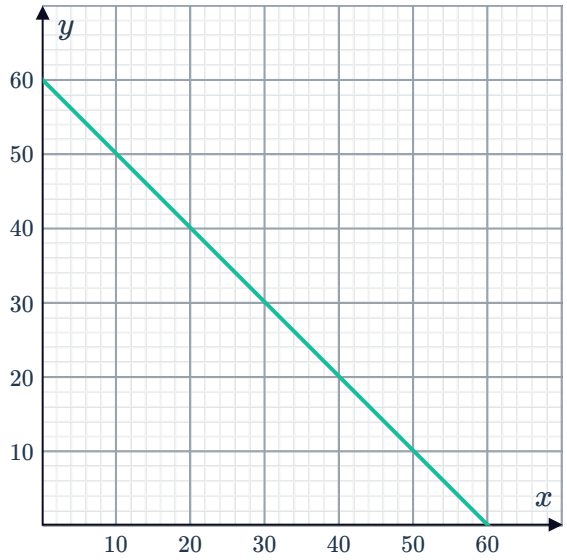
a

Number of minutes passed (x)	0	5	10	15	20	60
Amount of fuel left in tank (y)	60	55	50	45	40	0

b $y = 60 - x$

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c



29 $y = 20x + 110$

30

a $y = 30 - 5x$

b The race is 30 miles and he is running at 5 miles per hour, so he will finish the race in 6 hours. Time cannot be negative and he will not be running any more after 6 hours, so the restriction is $0 \leq x \leq 6$.

31

a $y = -2.1x + 63$

b The water level is falling at a constant rate of 2.1 feet per minute.

c The water level was 63 feet before any water was pumped out.

d 25.2 ft

e An appropriate domain is $x \in [0, 30]$ because solving the equation $0 = -2.1x + 63$ we get $x = 30$ when the well is emptied. Once the well is emptied, the context is no longer valid. Time is always positive, so we get $[0, 30]$.

32

a $y = 35x + 250$

b The total amount charged increases by $\$35$ for each additional hour of work.

c The minimum amount charged by the carpenter, which is the initial fee of $\$250$.

d The total cost is Materials+Labor. It will cost $\$1800 + 250 + 35 \times 12 = 2470$, and $2470 < 2500$, so she has budgeted enough for the ramp.

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33

a 1.63 ft/min

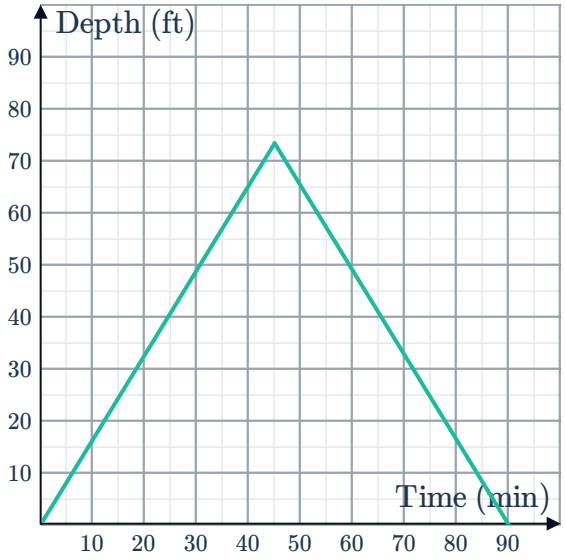
b $y = 1.63x$

c 9.78 ft

d $x = 7$

e The domain will be $x \in [0, 45]$ as after 45 minutes the diver will need to stop going deeper and start coming back to the surface.

f $y = -1.63(x - 45) + 73.35 = -1.63x + 146.7$ for $x \in [45, 90]$



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